



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**SENIOR CERTIFICATE EXAMINATIONS/
NATIONAL SENIOR CERTIFICATE EXAMINATIONS
SENIORSERTIFIKAAT-EKSAMEN/
NASIONALE SENIORSERTIFIKAAT-EKSAMEN**

**PHYSICAL SCIENCES: CHEMISTRY (P2)
FISIESE WETENSKAPPE: CHEMIE (V2)**

MAY/JUNE/MEI/JUNIE 2026

MARKING GUIDELINES/NASIEENRIGLYNE

MARKS/PUNTE: 150

**These marking guidelines consist of 16 pages./
Hierdie nasienriglyne bestaan uit 16 bladsye.**

QUESTION 1/VRAAG 1

- 1.1 A ✓✓ (2)
- 1.2 C ✓✓ (2)
- 1.3 C ✓✓ (2)
- 1.4 C ✓✓ (2)
- 1.5 D ✓✓ (2)
- 1.6 A ✓✓ (2)
- 1.7 A ✓✓ (2)
- 1.8 D ✓✓ (2)
- 1.9 B ✓✓ (2)
- 1.10 B ✓✓ (2)
- [20]**

QUESTION 2/VRAAG 2

- 2.1 Compounds in which there are no multiple bonds between C atoms in their hydrocarbon chains. ✓✓ **(2 or/of 0)**

A compound:

that has ONLY single bonds between C atoms.

that has no double OR triple bonds OR multiple bonds between C atoms.

Verbindings waarin daar geen meervoudige bindings tussen C-atome in hul koolwaterstofkettings is nie

Verbinding wat:

SLEGS enkelbindingsbevat tussen C-atome bevat.

geen dubbel- OF trippelbindings OF meervoudige bindings tussen C-atome nie.

(2)

- 2.2.1 E ✓ (1)
- 2.2.2 D ✓ (1)
- 2.2.3 A and/en F ✓ (1)

2.3.1

Marking criteria:

- Correct stem, i.e. heptyne. ✓
- Substituent (methyl) correctly identified. ✓
- IUPAC name completely correct including numbering, sequence, hyphens and commas. ✓

Nasienkriteria:

- *Korrekte stam, d.i. heptyn.* ✓
- *Substituente (metiel) korrek geïdentifiseer.* ✓
- *IUPAC-naam heeltemal korrek insluitende nommering, volgorde, koppeltekens en kommas.* ✓

4-methylhept-2-yne /4-methyl-2-heptyne ✓✓✓

4-metielhept-2-yn /4-metiel-2-heptyn

(3)

2.3.2

Marking criteria:

- Correct stem, i.e. hexane. ✓
- All substituents (dibromo and dimethyl) correctly identified. ✓
- IUPAC name completely correct including numbering, sequence, hyphens and commas. ✓

Nasienkriteria:

- *Korrekte stam d.i. heksaan.* ✓
- *Alle substituenten (dibromo en dimetiel) korrek geïdentifiseer.* ✓
- *IUPAC-naam heeltemal korrek insluitende nommering, volgorde, koppeltekens en kommas.* ✓

2,4-dibromo-3,5-dimethylhexane/2,4-dibromo-3,5-dimetielheksaan ✓✓✓

(3)

2.4.1

Marking criteria/Nasienkriteria

If any one of the underlined key phrases in the **correct context** is omitted, deduct 1 mark. /Indien enige van die onderstreepte frases in die **korrekte konteks** uitgelaat is, trek 1 punt af.

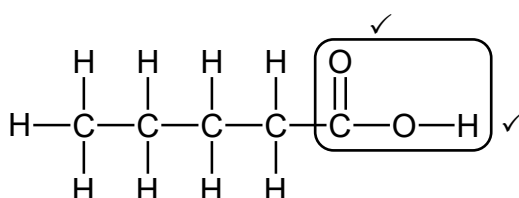
The underlined phrases must be in the correct context. / Die onderstreepte frases moet in die korrekte konteks wees.

Compounds with the same molecular formula, but different functional groups. ✓✓

Verbindings met dieselfde molekulêre formule, maar verskillende funksionele groepe.

(2)

2.4.2

**Marking criteria:**

- Functional group correct. ✓
- Whole structural formula correct. ✓

Nasienkriteria:

- *Funksionele groep korrek.* ✓
- *Hele struktuurformule korrek.* ✓

(2)

2.4.3

Marking criteria:

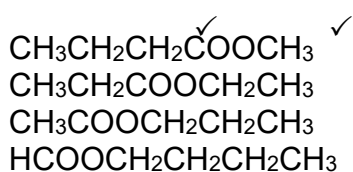
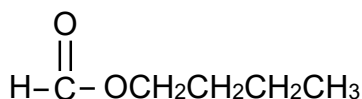
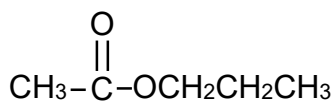
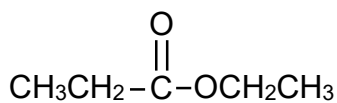
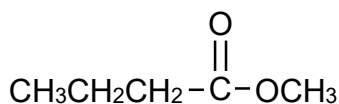
- Functional group correct. ✓
- Whole condensed structural formula correct. ✓

Nasienkriteria:

- *Funksionele groep korrek.* ✓
- *Hele gekondenseerde struktuurformule korrek.* ✓

NOTE/AANTEKENING:

- Structural formulae used/*Struktuurformule gebruik.* Max./Maks. $\frac{1}{2}$
- Molecular formulae used/*Molekulêre formule gebruik.* Max./Maks. $\frac{0}{2}$
- Accept bonds on functional group in condensed structure/*Aanvaar bindings in funksionele groep.*

**OR/OF**

(2)

2.5

Secondary/*Sekondêre* ✓

The halogen (-X/-Br) is bonded to a C atom that is bonded to two other C atoms. ✓

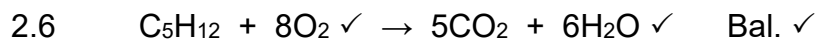
Die halogeen (-X/-Br) is gebind aan 'n C-atoom wat gebind is aan twee ander C-atome

OR

The functional group $\begin{array}{c} | \\ \text{--- C ---} \\ | \\ \text{X/Br} \end{array}$ is bonded to two other C atoms.

Die funksionele groep $\begin{array}{c} | \\ \text{--- C ---} \\ | \\ \text{X/Br} \end{array}$ is gebind aan twee ander C-atome.

(2)



Ignore phases./Ignoreer fases.

Marking criteria/Nasienkriteria:

- Reactants $\checkmark$ Products $\checkmark$ Balancing: $\checkmark$
Reaktanse Produkte Balansering
- Ignore double arrows./Ignoreer dubbelpyle.
- Marking rule 6.3.10/Nasienreël 6.3.10.

IF/INDIEN:

Structural formula for C_5H_{12} penalise 1 mark

Struktuur formule vir C_5H_{12} - penaliseer 1 punt.

Equal sign instead of arrow - penalise 1 mark.

Gelykaanteken ipv pyl – penaliseer 1 punt.

(3)
[22]

QUESTION 3/VRAAG 3

3.1.1 II $\checkmark$ (1)

3.1.2 II $\checkmark$ (1)

3.1.3 I $\checkmark$ (1)

3.2.1 **Marking criteria/Nasienkriteria**
 If any one of the underlined key phrases in the **correct context** is omitted, deduct 1 mark./Indien enige van die onderstreepte frases in die **korrekte konteks** uitgelaat is, trek 1 punt af.

The temperature at which the vapour pressure of a substance equals atmospheric pressure. $\checkmark\checkmark$

Die temperatuur waar die dampdruk van 'n stof gelyk is aan atmosferiese druk.

(2)

3.2.2 A, B and C are straight chain primary alcohols $\checkmark$
A, B en C is reguitketting primêre alkohole (1)

3.2.3 London/dispersion/Van de Waals forces $\checkmark$
London/dispersie/Van de Waalskragte (1)

3.2.4 Increase in chain length $\checkmark$
Toename in kettinglengte (1)

3.2.5

Marking criteria:

- Name the TYPES of intermolecular forces. ✓✓
- Compare the strength of intermolecular forces. ✓
- OR** Compare the energy required to overcome intermolecular forces.
- Compare the boiling points ✓

Nasienkriteria:

- Vergelyk die *TIPE* intermolekulêre kragte .
- Vergelyk die *sterkte* van intermolekulêre kragte.
- OF** Vergelyk die *energie* benodig om intermolekulêre kragte te oorkom.
- Vergelyk die kookpunt

- Between butan-1-ol molecules are hydrogen bonds (in addition to London and dipole-dipole forces). ✓
- Between butanal molecules are dipole-dipole forces (in addition to London forces). ✓
- Hydrogen bonds are stronger than dipole-dipole forces ✓/More energy needed to overcome/ break hydrogen bonds/ At given temperature less molecules will be in vapour phase.

OR

Dipole-dipole forces are weaker than hydrogen bonds /Less energy needed to overcome/ break dipole-dipole forces./ At given temperature more molecules will be in vapour phase.

- Butan-1-ol has a higher boiling point **OR** Butanal has lower boiling point ✓

- *Tussen die butan-1-ol molekules is waterstofbindings (asook London en dipool-dipoolkragte).*
- *Tussen die butanaal molekules is dipool-dipoolkragte (asook Londonkragte).*
- *Waterstofbindings is sterker as die dipool-dipoolkragte/ Meer energie is nodig om die waterstofbinding te oorkom/breek./ By 'n gegewe temperatuur is minder molekules in dampfase.*

OF

Dipool-dipoolkragte is swakker as die waterstofbinding./ Minder energie word benodig om die dipool-dipoolkragte te oorkom/ breek as die waterstofbindings. By 'n gegewe temperatuur is meer molekules in die dampfase.

- Butan-1-ol het 'n hoër kookpunt **OF** Butanaal het 'n laer kookpunt

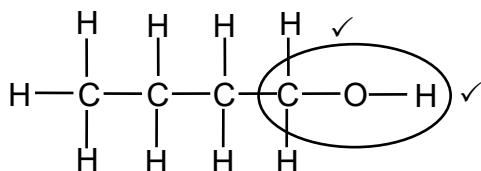
(4)
[12]

QUESTION 4/VRAAG 4

4.1.1 Substitution/halogenation/bromination ✓✓
Substitusie/halogenering/brominerig (2)

4.1.2 Heat/sunlight/UV / *Hitte/sonlig* ✓ (1)

4.2.1

**Marking criteria:**

- Functional group correct. ✓
- Whole structural formula correct. ✓

Nasienkriteria:

- *Funksionele groep korrek.* ✓
- *Hele struktuurformule korrek.* ✓

(2)

4.2.2 Esterification/Condensation/*Veresterig/Esterifikasie/Kondensasie* ✓ (1)

4.2.3 Butyl ✓ methanoate ✓ / *Butielmetanoaat* (2)

4.3 Substitution/Hydrolysis / *Substitusie/Hidrolise* ✓✓ (2)

4.4 Hydrogen bromide/HBr / *Waterstofbromied* ✓

ACCEPT/AANVAAR:

Sodium bromide AND sulphuric acid/NaBr + H₂SO₄

Natriumbromied EN swawelsuur

(1)

4.5.1 Elimination / *Eliminasie* ✓ (1)

4.5.2 Concentrated strong base/NaOH/KOH/LiOH/ Gekonstreerde sterk basis ✓ (1)

4.5.3 Concentrated sulphuric acid/H₂SO₄/ Gekonsentreerde swawelsuur/ ✓
Concentrated phosphoric acid/H₃PO₄/ Gekonsentreerde fosforsuur (1)

4.5.4

Marking criteria:

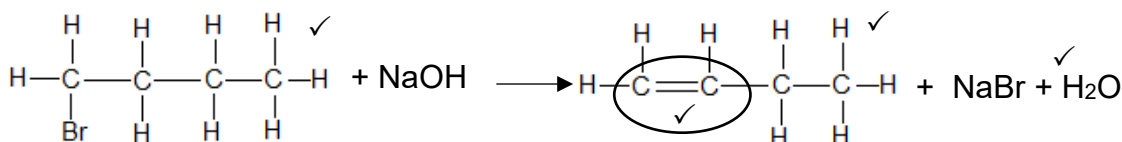
- Whole structural formula of bromoalkane correct (functional group on first C atom). ✓
- Functional group of alkene correct. ✓
- Whole structural formula of alkene correct (functional group on first C atom). ✓
- NaOH , $\text{NaBr} + \text{H}_2\text{O}$ / LiOH , $\text{LiBr} + \text{H}_2\text{O}$ / KOH , $\text{KBr} + \text{H}_2\text{O}$ ✓

Nasienkriteria:

- Hele struktuurformule van van haloakaan korrek (funksionele groep op de eerste C-atoom). ✓
- Funksionele groep van alkeen korrek.
- Hele struktuurformule van van alkeen korrek (funksionele groep op de eerste C-atoom). ✓
- NaOH , $\text{NaBr} + \text{H}_2\text{O}$ / LiOH , $\text{LiBr} + \text{H}_2\text{O}$ / KOH , $\text{KBr} + \text{H}_2\text{O}$ ✓

IF/INDIEN

- Condensed, semi structural or molecular formula.
Gekondenseerde, semi-struktuurformule of molekulêre formule. Max/Maks: $\frac{1}{5}$
- Marking rule 6.3.10/Nasienreël 6.3.10

(4)
[17]**QUESTION 5/VRAAG 5**

- 5.1 A substance where the number of moles is stoichiometrically more than needed / A substance that is not first consumed completely /used up in a chemical reaction. ✓✓
Die stof waarvan die aantal mol stoichiometries meer is as wat benodig word.
/Die stof wat nie eerste opgebruik is nie in 'n chemiese reaksie. (2)

- 5.2.1
- At t_3 , the concentration of HNO_3 has decreased/Cu is being used up. ✓
 - Less (HNO_3) particles per unit volume. ✓
 - Less effective collisions per unit time./Lower frequency of effective collisions. ✓
 - By t_3 , het die konsentrasie van HNO_3 afgeneem/Cu word opgebruik.
 - Minder (HNO_3) deeltjies per eenheidsvolume.
 - Minder effektiewe botsings per eenheidtyd/sekonde. / Laer frekwensie van effektiewe botsings. (3)

- 5.2.2 Higher than / Hoër as ✓ (1)

- 5.2.3 Steeper gradient ✓
Steiler helling (1)

5.2.4

Marking criteria/Nasienkriteria:

- Substitute the volume and rate into rate formula. ✓
Vervang volume en die tempo in tempo formule.
- Final correct answer/Finale korrekte antwoord: 85 s ✓

$$\text{Ave rate/gem tempo} = \frac{\Delta V(\text{NO}_2)}{\Delta t}$$

$$0,02 = \frac{1,7}{t_2} \quad \checkmark$$

$$t_2 = 85 \text{ (s)} \quad \checkmark$$

(2)

5.2.5

Marking criteria

- (a) Formula $n = \frac{V}{V_m}$ or $n = cV$ ✓
 (b) Substitute 12 and 0,04 in $n = cV$ ✓
 (c) Substitute 25 in $n = \frac{V}{V_m}$ ✓
 (d) USE mol ratio:
 $n(\text{HNO}_3) : n(\text{NO}_2) = 4 : 2$ ✓
 (e) Calculate excess moles
 $n(\text{HNO}_3)_{\text{initial}} - n(\text{HNO}_3)_{\text{used}}$ /
 $c(\text{HNO}_3)_{\text{initial}} - c(\text{HNO}_3)_{\text{used}}$. ✓
 (f) Final correct answer: $4,5 \text{ mol} \cdot \text{dm}^{-3}$ ✓
 Range: 4 to $4,5 \text{ mol} \cdot \text{dm}^{-3}$

Nasienkriteria:

- (a) Formule $n = \frac{V}{V_m}$ or $n = CV$ ✓
 (b) Vervang 12 en 0,04 in $n = cV$ ✓
 (c) Vervang 25 in $n = \frac{V}{V_m}$ ✓
 (d) GEBRUIK molverhouding:
 $n(\text{HNO}_3) : n(\text{NO}_2) = 4 : 2$ ✓
 a) Bereken oormaat n
 $(\text{HNO}_3)_{\text{aanvank}} - n(\text{HNO}_3)_{\text{gebruik}}$
 $c(\text{HNO}_3)_{\text{aanvank}} - c(\text{HNO}_3)_{\text{gebruik}}$. ✓
 b) Finale korrekte antwoord:
 $4,5 \text{ mol} \cdot \text{dm}^{-3}$ ✓
 Gebied: 4 to $4,5 \text{ mol} \cdot \text{dm}^{-3}$

$$\begin{aligned} n(\text{HNO}_3)_{\text{ini}} &= cV \quad \checkmark \\ &= (12)(0,02) \quad \checkmark \\ &= 0,24 \text{ mol} \end{aligned}$$

$$\begin{aligned} n(\text{NO}_2)_{\text{prod}} &= \frac{V}{V_m} \\ &= \frac{1,92}{25} \quad \checkmark \\ &= 0,077 \text{ mol} \end{aligned}$$

$$\begin{aligned} n(\text{HNO}_3)_{\text{used}} &= \frac{4}{2} n(\text{NO}_2) \\ &= \frac{4}{2} (0,077) \quad \checkmark \\ &= 0,15 \text{ mol} \end{aligned}$$

$$\begin{aligned} n(\text{HNO}_3)_{\text{excess}} &= 0,24 - 0,15 \quad \checkmark \\ &= 0,09 \text{ mol} \end{aligned}$$

$$\begin{aligned} c(\text{HNO}_3)_{\text{excess}} &= \frac{n}{V} \\ &= \frac{0,09}{0,02} \\ &= 4,32 \text{ mol} \cdot \text{dm}^{-3} \end{aligned}$$

$$c(\text{H}_3\text{O}^+) = c(\text{HNO}_3) = 4,32 \text{ mol} \cdot \text{dm}^{-3} \quad \checkmark$$

(6)

- 5.2.6 B ✓✓ (2)
- 5.3 (+)5 ✓✓ (2)
- [19]

QUESTION 6/VRAAG 6

6.1

Marking criteria/Nasienkriteria

If any one of the underlined key phrases in the **correct context** is omitted, deduct 1 mark./Indien enige van die onderstreepte frases in die **korrekte konteks** uitgelaat is, trek 1 punt af.

When the equilibrium in a closed system is disturbed, the system will re-instate a new equilibrium by favouring the reaction that will cancel/oppose the disturbance. ✓✓

Wanneer die ewewig in 'n geslote sisteem versteur word, sal die sisteem 'n nuwe ewewig instel deur die reaksie te bevoordeel wat die versteuring kanselleer/teenwerk. (2)

- 6.2 Equilibrium is reached. / Ewewig is bereik. ✓ (1)

6.3 Decreased / Verlaag ✓

- Forward reaction is favoured/ Concentration of ICl_3 /products increases/ Concentration of I_2 and Cl_2 /reactants decreases. ✓
- The reaction that produces a fewer number of moles/amount of gas (4 mole gas to 2 moles gas) is favoured/The reaction that decreases the pressure is favoured. ✓

- Voorwaartse reaksie is bevoordeel/Konsentrasie van ICl_3 /produkte verhoog /Konsentrasie van I_2 en Cl_2 /reaktanse verminder.
- Die reaksie wat 'n kleiner aantal mol/hoeveelheid gas (4 mol gas na 2 mol gas) produseer word bevoordeel./Die reaksie wat die druk verlaag word bevoordeel. (3)

- 6.4 Decrease in temperature./Verlaging van temperatuur. ✓✓

IF/INDIEN:

Temperature/Temperatuur

Max/Maks: $\frac{1}{2}$

(2)

6.5

CALCULATIONS USING CONCENTRATION**BEREKENINGE WAT KONSENTRASIE GEBRUIK****Marking criteria:**

- (a) Initial concentrations. ✓
 (b) **USE** ratio: $I_2(g): 3Cl_2(g): 2ICl_3(g)$ ✓
 (c) Equilibrium $[I_2] = \text{initial } [I_2] - \text{change in } [I_2]$
 Equilibrium $[Cl_2] = \text{initial } [Cl_2] - \text{change in } [Cl_2]$
 Equilibrium $[ICl_3] = \text{initial } [ICl_3] + \text{change in } [ICl_3]$ } ✓
 (d) K_c expression ✓
 (e) Substitute K_c value in K_c expression ✓
 (f) Substitute concentrations in K_c expression ✓
 (g) Calculate $n(I_2)$, substitute 2 dm^3 in $n = cV$ ✓
 (h) **CORRECT** final answer. ✓
 Range: $1,34 - 1,42 \text{ mol}$

Nasienkriteria:

- (a) Aanvangskonsentrasie. ✓
 (b) **GEBRUIK** verhouding: $I_2(g): 3Cl_2(g): 2ICl_3(g)$ ✓
 (c) Ewewig $[I_2] = \text{initial } [I_2] - \text{change in } [I_2]$
 Equilibrium $[Cl_2] = \text{initial } [Cl_2] - \text{change in } [Cl_2]$
 Equilibrium $[ICl_3] = \text{initial } [ICl_3] + \text{change in } [ICl_3]$ } ✓
 (d) K_c -uitdrukking ✓
 (e) Vervang K_c waarde in K_c -uitdrukking ✓
 (f) Vervang konsentrasies in K_c -uitdrukking ✓
 (g) Bereken $n(I_2)$, vervang 2 dm^3 in $n = cV$ ✓
 (h) **KORREKTE** finale antwoord. ✓
Gebied: $1,34 - 1,42 \text{ mol}$

OPTION 1/OPSIE 1

	$I_2(g)$	$Cl_2(g)$	$ICl_3(g)$
Initial concentration ($\text{mol} \cdot \text{dm}^{-3}$) <i>Aanvangskonsentrasie ($\text{mol} \cdot \text{dm}^{-3}$)</i>	$0,15 + x$	$0,36$	$0,05$
Change in concentration ($\text{mol} \cdot \text{dm}^{-3}$) <i>Verandering in konsentrasie ($\text{mol} \cdot \text{dm}^{-3}$)</i>	$0,02$	$0,06$	$0,04$
Equilibrium concentration ($\text{mol} \cdot \text{dm}^{-3}$) <i>Ewewigskonsentrasie ($\text{mol} \cdot \text{dm}^{-3}$)</i>	$0,13 + x$	$0,3$	$0,09$

✓ (a)

✓ (b)

✓ (c)

$$K_c = \frac{[ICl_3]^2}{[I_2][Cl_2]^3} \quad \checkmark (d)$$

$$0,357 \quad \checkmark (e) = \frac{(0,09)^2}{\left(\frac{0,26 + x}{2}\right)(0,3)^3} \quad \checkmark (f)$$

No K_c expression, correct substitution/Geen K_c -uitdrukking, korrekte substitusie: Max./Maks. $\frac{7}{8}$

Wrong K_c expression/
Verkeerde K_c -uitdrukking: Max./Maks. $\frac{5}{8}$

$$x = 0,7103 \text{ mol} \cdot \text{dm}^{-3}$$

$$n I_2(g) = cV \quad \checkmark (g)$$

$$= (0,7103)(2) \quad \checkmark (h)$$

$$= 1,4206 \text{ mol}$$

OPTION 2/OPSIE 2

	I ₂ (g)	Cl ₂ (g)	ICl ₃ (g)
Initial moles (mol) <i>Aanvangsmol (mol·dm⁻³)</i>	0,3 + x	0,72	0,1
Change in moles (mol) <i>Verandering in mol (mol·dm⁻³)</i>	0,04	0,12	0,08
Equilibrium moles (mol) <i>Ewewigsmol (mol·dm⁻³)</i>	0,26 + x	0,6	0,18
Equilibrium concentration (mol·dm ⁻³) <i>Ewewigskonsentrasie (mol·dm⁻³)</i>	$\frac{0,26 + x}{2}$	0,3	0,09

✓(a)

✓(b)

✓(c)

✓(g)

$$K_c = \frac{[ICl_3]^2}{[I_2][Cl_2]^3} \quad \checkmark (d)$$

$$0,357 \quad \checkmark (e) = \frac{(0,09)^2}{\left(\frac{0,26 + x}{2}\right)(0,3)^3} \quad \checkmark (f)$$

$$x = 1,4206 \text{ mol} \quad \checkmark (h)$$

No K_c expression, correct substitution/*Geen K_c-uitdrukking, korrekte substitusie: Max./Maks. 7/8*

Wrong K_c expression/
Verkeerde K_c-uitdrukking: Max./Maks. 5/8

(8)
[16]**QUESTION 7/VRAAG 7**

- 7.1.1
- Endothermic ✓
 - Forward reaction is favoured. ✓/K_w increased with an increase in temperature.
 - Increase in temperature favours endothermic reaction. ✓
 - *Endotermies*
 - *Voorwaartse reaksie word bevoordeel./K_w verhoog met 'n toename in temperatuur.*
 - *Verhoging in temperatuur bevoordeel die endotermiese reaksie.*

(3)

7.1.2

OPTION 1/OPSIE 1

$$K_w = [H_3O^+][OH^-]$$

$$5,48 \times 10^{-14} = [H_3O^+]^2 \quad \checkmark$$

$$[H_3O^+] = 2,34 \times 10^{-7}$$

$$pH = -\log[H_3O^+] \quad \checkmark$$

$$= -\log(2,34 \times 10^{-7}) \quad \checkmark$$

$$= 6,63 \quad \checkmark$$

OPTION 2/OPSIE 2

$$pH = -\log[H_3O^+] \quad \checkmark$$

$$= -\log \sqrt{5,48 \times 10^{-14}} \quad \checkmark \checkmark$$

$$= 6,63 \quad \checkmark$$

(4)

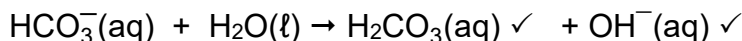
7.2.1

Marking criteria/Nasienkriteria:Products / *Produkte* ✓✓ one mark for each/een punt vir elk

Marking rule 6.3.10/Nasienreël 6.3.10

IF/INDIEN:Ignore/*Ignoreer* ⇒ and phases/*en fases*

Equal sign instead of ⇒ ignore/ Gelykaanteken ipv ⇒ ignoreer



(2)

7.2.2

Marking criteria/Nasienkriteria:Products / *Produkte* ✓✓ one mark for each/een punt vir elk

Marking rule 6.3.10/Nasienreël 6.3.10

IF/INDIEN:Ignore/*Ignoreer* ⇒ and phases/*en fases*

Equal sign instead of ⇒ ignore/ Gelykaanteken ipv ⇒ ignoreer



(2)

NOTE:

If Q7.2.1 and 7.2.2 are swapped around, then 2/4 max.

LET WEL:*Indien V7.2.1 en 7.2.2 omgeruil word, dan 2/4 maks.*

7.3.1

ANY ONE:

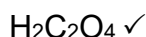
- A substance whose (aqueous) solution contains ions. ✓
- Substance that dissolves in water to give a solution that conducts electricity.
- A substance that forms ions in water / when melted.

ENIGE EEN:

- 'n Stof waarvan die (waterige) oplossing ione bevat. ✓
- 'n Stof wat in water oplos om 'n oplossing te vorm wat elektrisiteit gelei.
- 'n Stof wat ione in water vorm / wanneer gesmelt word.

(1)

7.3.2

Highest K_a value/Strongest acid/Highest concentration of ions formed. ✓*Hoogste K_a waarde/Sterkste suur./Hoogste konsentrasie ione vorm.*

(2)

7.4.1

Marking criteria	Nasienkriteria:
(a) Substitute: $0,5 \text{ mol} \cdot \text{dm}^{-3}$ & $0,05 \text{ dm}^3$ in $n = cV$ ✓	(a) Vervang: $0,5 \text{ mol} \cdot \text{dm}^{-3}$ & $0,05 \text{ dm}^3$ in $n = cV$ ✓
(b) Substitute: $0,378 \text{ mol} \cdot \text{dm}^{-3}$ & $0,0265 \text{ dm}^3$ in $n = cV$ ✓	(b) Vervang: $0,378 \text{ mol} \cdot \text{dm}^{-3}$ & $0,0265 \text{ dm}^3$ in $n = cV$ ✓
(c) USE ratio: $n(\text{HCl}) = n(\text{NaOH})$ ✓	(c) GEBRUIK verhouding ratio: $n(\text{HCl}) = n(\text{NaOH})$ ✓
(d) $n\text{HCl} = n_{\text{ini}} - n_{\text{excess}}$ ✓	(d) $n\text{HCl} = n_{\text{aanvanklik}} - n_{\text{oormaat}}$ ✓
(e) Final answer: $0,015 \text{ mol}$ ✓ Range: $0,01498$ to $0,02 \text{ mol}$	(e) Finale antwoord: $0,015 \text{ mol}$ ✓ Gebied: $0,01498$ tot $0,02 \text{ mol}$

$$n\text{HCl} = cV$$

$$= (0,5)(0,05) \checkmark \text{ (a)}$$

$$= 0,025 \text{ mol}$$

$$n\text{NaOH in titrate} = cV$$

$$= (0,378)(0,0265) \checkmark \text{ (b)}$$

$$= 1 \times 10^{-2} \text{ mol}$$

$$n\text{HCl} = n\text{NaOH} \checkmark \text{ (c)}$$

$$= 1 \times 10^{-2} \text{ mol}$$

$$n\text{HCl reacted with } n\text{OH}^- \text{ in tablet} = 0,025 - 1 \times 10^{-2} \checkmark \text{ (d)}$$

$$= 0,015 \text{ mol}$$

$$n\text{OH}^- = n\text{HCl} \downarrow$$

$$= 0,015 \text{ mol} \checkmark \text{ (e)}$$

(5)

7.4.2

POSITIVE MARKING FROM QUESTION 7.4.1/
POSITIEWE NASIEN VANAF VRAAG 7.4.1

Marking criteria	Nasienkriteria:
(a) 5 parts of $n\text{OH}^-$ ions in tablet ✓	(a) 5 dele van $n\text{OH}^-$ ione in tablet ✓
(b) 3 parts from $n\text{Al}(\text{OH})_3$ ✓	(b) 3 dele van $n\text{Al}(\text{OH})_3$ ✓
(c) Use ratio $n\text{Al}(\text{OH})_3 : n\text{OH}^-$ ✓	(c) Gebruik verhouding $n\text{Al}(\text{OH})_3 : n\text{OH}^-$ ✓
(d) Substitute: $n\text{Al}(\text{OH})_3$ and 78 in $n = \frac{m}{M}$ ✓	(d) Vervang: $n\text{Al}(\text{OH})_3$ en 78 in $n = \frac{m}{M}$ ✓
(e) Final answer = $0,234 \text{ g}$ ✓	(e) Finale antwoord = $0,234 \text{ g}$ ✓

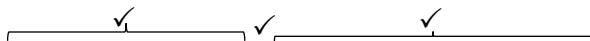
OPTION 1/OPSIE 1	OPTION 2/OPSIE 2
$n\text{OH}^- = 3x + 2x \checkmark \text{ (a)} = 0,015 \text{ mol}$ $n\text{OH}^- \text{ from/van } \text{Al}(\text{OH})_3$ $= \frac{3x}{5x} \times 0,015 \checkmark \text{ (b)}$ $= 9 \times 10^{-3} \text{ mol}$ $n\text{Al}(\text{OH})_3 = \frac{9 \times 10^{-3}}{3} \checkmark \text{ (c)}$ $= 3 \times 10^{-3} \text{ mol Al}(\text{OH})_3$ $n\text{Al}(\text{OH})_3 = \frac{m}{M}$ $3 \times 10^{-3} = \frac{m}{78} \checkmark \text{ (d)}$ $m = 0,234 \text{ g} \checkmark \text{ (e)}$	$\frac{3}{5} \times 0,015 \text{ OH}^- = n\text{Al}(\text{OH})_3 \checkmark \text{ (b)}$ $= 9 \times 10^{-3} \checkmark \text{ (a)}$ $n\text{Al}(\text{OH})_3 = \frac{9 \times 10^{-3}}{3} \checkmark \text{ (c)}$ $= 3 \times 10^{-3} \text{ mol Al}(\text{OH})_3$ $n\text{Al}(\text{OH})_3 = \frac{m}{M}$ $3 \times 10^{-3} = \frac{m}{78} \checkmark \text{ (d)}$ $m = 0,234 \text{ g} \checkmark \text{ (e)}$

(5)

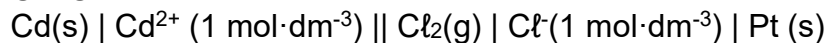
[24]

QUESTION 8/VRAAG 8

- 8.1 A cell in which chemical energy is converted to electrical energy. ✓✓
 'n Sel waarin *chemiese energie omgeskakel word na elektriese energie.* (2)



- 8.2.1 $\text{Cd(s)} \mid \text{Cd}^{2+}(\text{aq}) \parallel \text{Cl}_2(\text{g}) \mid \text{Cl}^-(\text{aq}) \mid \text{Pt(s)}$

OR/OF**OR/OF**

- 8.2.2 Q to/na P ✓ (1)

- 8.2.3 **Marking criteria/Nasienkriteria:**

- $\text{Cl}_2(\text{g}) + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-(\text{aq}) \quad (1/2) \quad 2\text{Cl}^-(\text{aq}) \leftarrow \text{Cl}_2(\text{g}) + 2\text{e}^- \quad (2/2)$
- $\text{Cl}_2(\text{g}) + 2\text{e}^- \leftarrow 2\text{Cl}^-(\text{aq}) \quad (0/2) \quad 2\text{Cl}^-(\text{aq}) \rightleftharpoons \text{Cl}_2(\text{g}) + 2\text{e}^- \quad (0/2)$
- Ignore if charge omitted on electron. / *Ignoreer indien lading weggelaat op elektron.*
- If charge (-) omitted on Cl^- / *Indien lading (-) weggelaat op Cl^- .*
 Example/Voorbeeld: $\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}$ Max./Maks: $1/2$
- Ignore phases / *Ignoreer fases.*



- 8.3 Decreases. / *Verminder.* ✓
 Reverse reaction is favoured. / *Terugwaartse reaksie is bevoordeel.* ✓ (2)

[10]

QUESTION 9/VRAAG 9

- 9.1 The chemical process in which electrical energy is converted to chemical energy. ✓✓

OR

The use of electrical energy to produce a chemical change.

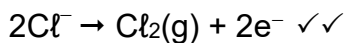
Die chemiese proses waar elektriese energie omgeskakel word in chemiese energie.

OF

Die gebruik van elektriese energie om 'n chemiese verandering te weeg te bring.

(2)

- 9.2 $2\text{H}_2\text{O} \rightarrow \text{O}_2(\text{g}) + 4\text{H}^+ + 4\text{e}^-$ ✓✓



(4)

- 9.3 Cl_2 /Chlorine/*Chloor* ✓

(1)

- 9.4.1 Copper / Cu / *Koper* ✓

(1)

- 9.4.2 Iodine / I_2 / *Jodium* ✓

(1)

[9]

TOTAL/TOTAAL: 150